

Third-Party Trackers in COVID-19 Mobile Applications Can Enable Privacy Leaks

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ABSTRACT

Since 2019, the world has been facing an unprecedented pandemic. Governments and various organizations have made great efforts to develop applications that aid in controlling the spread of COVID-19 and improve people's quality of life. However, our research has discovered that many of these applications contain third-party libraries, including trackers, which can impact user privacy. Out of the 595 applications we reviewed, 402 included at least one tracker, and sensitive information was being transferred to these tracker servers. From a privacy perspective, including trackers in sensitive applications must be reconsidered, primarily when governments sponsor them.

CCS CONCEPTS

• Security and privacy → Software and application security.

KEYWORDS

Privacy, Android, SDK, COVID-19

ACM Reference Format:

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1 INTRODUCTION

We are all familiar with Coronavirus (COVID-19), an infectious disease caused by the SARS-CoV-2 virus. Infected individuals typically experience mild to moderate respiratory illness [4]. The first case was reported towards the end of 2019, and it quickly spread to become a global pandemic. As of writing this, nearly 700 million infections and approximately 6.8 million deaths have been attributed to this virus [6]. While the weekly death rate has decreased compared to the first two years of the outbreak, COVID-19 remains a significant global health concern.

Technology has become an integral part of our daily lives, with digital health being a field that has experienced significant growth in recent years. This category encompasses various technologies

such as mobile health, health information technology, wearable devices, telehealth and telemedicine, and personalized medicine. In light of the COVID-19 pandemic, mobile applications for digital health have been developed and utilized globally. Governments, international organizations, health institutions, and universities are among the sponsors of this technology. Our focus has been studying mobile apps developed, provided, or sponsored by these entities. Please see Table 1 for a list of analyzed apps.

Users trust institutions sponsoring certain apps, like mandatory government apps, for accessing public places. However, COVID-19 contact tracing apps raise privacy concerns. Developers must comply with data protection laws and conduct impact assessments. Privacy policies and user consent are critical. Open-source code can contribute to transparency. Privacy and transparency must be balanced.

Mobile applications commonly incorporate third-party libraries using software development kits (SDKs). They offer numerous benefits to developers, such as particular functionalities like maps or login with social media credentials, monetization, crash reports, and user engagement. However, some libraries can obtain sensitive data from the user, their device, and their interaction, including location, device information, application attributes, and battery status. This data is then utilized to create user profiles to provide targeted advertisements.

Regarding gathering user information, tracker libraries may see value in every data acquired. However, for processors handling sensitive COVID-19-related information, a great deal of responsibility is required. As a result, any data collected and shared with external parties must be kept to a minimum and only utilized for the specific purposes of the application. Several countries recognized this need while developing their contact tracing applications, as outlined in [5]: "13. It is important to clearly define the use of coding libraries, frameworks, APIs, SDKs, and other software components, including those within the mobile operating system. To avoid data collection by third parties for unrelated purposes."

Research Questions. We examined over 500 COVID-19-related Android apps from reputable sources for the presence of trackers. We analyzed the information collected and assessed the impact on users. Our study answers these questions:

- Are trackers used?
- What information is tracked?
- What are the potential user impacts?

Contributions. Our research focused on analyzing the impact of trackers in COVID-19 apps, an area that has not been thoroughly explored in previous studies. Our findings indicate that a significant amount of sensitive information have been shared with the

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Country	Name	Developer	Goal
Australia	Coronavirus Australia	Australian Department of Health	Official Information
Brazil	Coronavirus - SUS	Ministério da Saúde	COVID Guidance (*)
Denmark	Coronapas	Danish Ministry of Health	COVID Passport
France	TousAntiCovid	Ministère de la Santé et de la Prévention	Contact Tracing (*)
Germany	Corona-Datenspende	Robert Koch-Institut	Studies
Hong Kong	StayHomeSafe	Office of the Gvment. Chief Inf. Officer	Quarantine Enforcement
India	West Bengal Emergency Fund	Govt. of West Bengal	Donations
International	COVID-19: response	United Nations	Information
Italy	LAZIOdrCovid	Salute Lazio	Telemedicine
Jordan	Cradar	Nat. Center for Sec. and Crisis Mgmt.	Gatherings Denouncing
Malaysia	MySejahtera	Government of Malaysia	Self-Diagnostic (*)
New Zealand	Āwhina - for health workers	Ministry of Health	Health Workers Support
Norway	Kontroll av koronasertifikat	Institute of Public Health	Certificates Verification
UAE	COVID-19 EHS	Emirates Health Services	Vaccination (*)
Uruguay	Coronavirus UY	E-Government Agency	Statistics (*)
US	COVID Coach	US Department of Veterans Affairs	General Well-Being
Vietnam	Vietnam Health Declaration	Ministry of Health	Travelers Health Declaration

Table 1: Example of analyzed apps and their main purposes. (*) These apps have other purposes besides those stated in the table.

servers of most trackers, including information that can potentially be used to infer health and COVID-19 related details of the user. Our research involved analyzing nearly 600 Android apps, a much larger sample size than previous studies, making our findings representative of the COVID-19 apps sponsored by governments and high-profile organizations. Additionally, we discuss the different privacy-oriented initiatives governments take, highlighting the discrepancies in app privacy policies and Data Protection Impact Assessments, where trackers are sometimes not disclosed or included.

Structure of the paper. The rest of the paper is structured as follows: Section 2 discusses related work. In Section 3 we describe the methodology used in our investigation, and in Section 4, we put forward the results of our research, provide details of the analyzed trackers and discuss the privacy issues caused by those tools. Section 5 elaborates on our findings and assesses their impact on violating data protection principles. Section 6 concludes and discusses further work.

2 RELATED WORK

We have come across various research papers on COVID-19 apps from different angles.

Numerous academic research studies and industry analyses have been conducted on privacy concerns in various apps since the onset of the pandemic [22, 29, 33, 40]. However, most of these studies did not focus on the impact of including SDKs in these apps, which could be risky as data privacy issues and sharing of personal information with third parties may remain unnoticed. The few studies that have discussed the use of trackers have mainly focused on identifying the specific trackers employed [21, 38]. While application permissions have received significant attention, it is not necessarily alarming for an app that uses *Bluetooth* for contact tracing to request Bluetooth-related permissions or for an app that facilitates video calls with a doctor to request *Camera* permissions. Finally,

Dehay et al. [28] demonstrated how third-party libraries can easily access private data of a device through dynamic analysis of apps, and linked this to the possibility of using contact tracing apps for COVID-19 containment. Our study, utilizing static analysis of the trackers' code, supports their findings and fills the gap in other studies by providing a more detailed analysis of SDKs and their data tracking, thus offering essential insights on privacy concerns in COVID-19 apps.

In addition to the impact of COVID-19 on privacy, the relationship between apps, trackers, and privacy has also been examined. Various tools have been developed to detect privacy breaches at the application level, such as AGRIGENTO[27], which outperformed previous tools. A dynamic privacy leakage analysis for third-parties trackers is presented in [30]. We chose to conduct a static analysis of the code of the selected apps using reverse engineering techniques and tools to determine the exact in-app library leaking private information and minimize false positives and negatives. Similar to our findings, the ecosystem of tracking companies was studied by several authors [24, 35], with Alphabet, Facebook, and Microsoft being among the most used Advertising and Tracking Services (ATS).

Additionally, other ATS highlighted in these studies were also found in our analysis (AppsFlyer, AdColony, Adobe, Appnext, ComScore, Flurry, Lotame, MixPanel, New Relic, Segment, and Startapp). In [32] the results of a privacy risk analysis focusing on analytic libraries for Android (instead of advertisement libraries, where most research has been done), show that apps also leak private information through these types of libraries. Our research found that analytic libraries for Android also pose a risk of privacy breaches, and we validated the tracking behavior of 10 out of 26 analytic libraries flagged as trackers by *Exodus* that effectively harvested and exfiltrated user and mobile information. The same was found in advertisement libraries for Android as reported in [36] and several

other studies. We also found tracking behaviour in two out of three ad libraries flagged as trackers.

The lack of consent from users for data gathering by third-party trackers is analyzed in [31]. Our study also revealed that the companies behind trackers do not always declare the actual data being collected, and any potential consent from users would still be inappropriate for the actual data harvesting being done by tracking libraries.

In [25] it is proposed *MobHide*, a data anonymization tool that could mitigate the findings of our research. A study conducted on mobile health apps, including what the trackers track, is reported in [37]. Our study showed that COVID-19 apps, as a subset of mobile health apps, shared the same lack of protection regarding sharing private information with third-parties.

3 METHODOLOGY

This section describes and discusses methodological aspects that guided our research.

3.1 Tracker detection

In this section, we will outline the steps taken to identify any tracking mechanisms present in COVID-19-related applications and the information they collect from the user’s device, app, and activity. Figure 1 illustrates the methodology approach that was utilized.

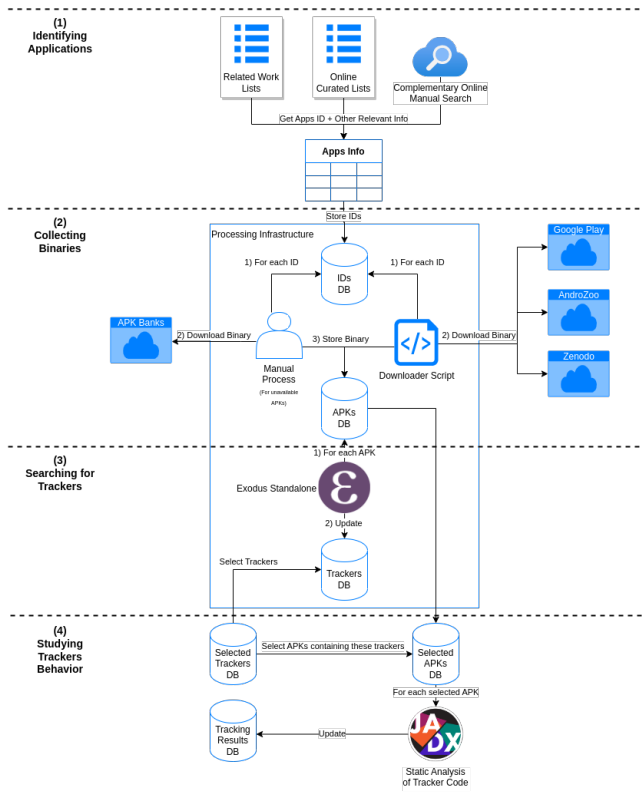


Figure 1: Tracker analysis methodology.

The term "tracker" is commonly used in related works to refer to a third-party library that is included in apps. This library provides

functionalities such as analytics, advertisements, and technical features without developers needing to code them. The library gathers information about the application, the device, and usage and then sends it to its servers.

3.1.1 Identifying Applications. Our initial objective was to locate COVID-19-related apps developed or sponsored by government entities, renowned international organizations, or reputable universities. We aimed to cover all relevant countries, regions, and independent jurisdictions for impartiality and thoroughness. This task was conducted between March 1st and March 15th, 2022.

To begin our search, we referred to related works, such as [26], and examined curated online lists, like [12]. We then supplemented these sources with our systematic online search to identify applications for all applicable countries and regions.

We took special care to verify the apps included in our study. We manually confirmed that each selected app was utilized in the context of the COVID-19 pandemic, unlike other automated crawling methods that may use selected keywords, such as "corona," leading to irrelevant results.

In total, we discovered 619 Android apps. Based on the Google Play download statistics, 80 apps were downloaded between 1 and 5 million times, 14 between 5 and 10 million times, 21 between 10 and 50 million times; one downloaded in the 50-100 million range and another over 100 million times.

3.1.2 Collecting Binaries. We gathered APK files using the Cluster UY infrastructure [34] for their collection, processing, and storage. Our primary source was the Google Play Store, and we downloaded as many binaries as possible from that source to maintain file integrity. However, certain apps were unavailable for download in our region or had been removed from the store, so we turned to three additional sources. Specifically, we accessed AndroZoo, which is a collection of Android apps maintained by Université du Luxembourg [23], and utilized a data set created by Wang et al. [39] for a previous study of malware in COVID-19 apps, which was hosted in the open data repository Zenodo [1]. Additionally, some binaries were not accessible through these sources, so we manually downloaded them from APK banks like Apksos, Apkpure, and Apkgk. We collected 595 binaries between March 15th and March 30th, 356 of which were from Google Play, 152 from AndroZoo, 8 from Zenodo, and 79 from manual downloads. For a comprehensive list of collected binaries, please refer to the list we made public [19].

3.1.3 Searching for Trackers. We utilized the *exodus-standalone* tool [16], to identify trackers within the collected binaries. This tool takes an APK file as input and generates a report that includes any trackers present in the binary. The tool detects trackers by performing signature matching. It extracts the Java classes from the APK file and compares their package names with a database of known trackers. We automated creating our tracker data set by setting the tool output as JSON. Our analysis revealed a total of 58 trackers present in the examined apps.

3.1.4 Studying Trackers Behavior. Extensive research was conducted to thoroughly investigate the data collected by trackers in the COVID-19 app ecosystem. The *JADX* tool for reverse engineering and static analyses [17] was utilized to accomplish this task. By analyzing the data flow through the tracker package, we

could pinpoint the exact classes in the APK file where data was collected and how this data was transmitted to their servers.

We randomly analyzed 22 out of the 58 trackers found in the apps, with the selection based on services provided, such as analytics, ads, push notifications, and others. As a result of this careful study, patterns were identified, starting from specific cases and converging to general observed practices. For more details, please refer to Section 4.

3.2 Limitations and research ethics

Our research encountered limitations, such as the absence of automated tools to perform behavioral analysis on Android apps giving precise and trustworthy results. As a result, we had to conduct most of the inspections manually. It is worth noting that we did not study all the trackers flagged by *Exodus* as it was beyond the scope of our investigation.

Just because a tracker is present in the application binary does not necessarily mean it will be executed. In our study, we assumed that if a tracker was present in the code, it would be run and the data collected, otherwise, that portion of code should not have been included in the released binary. During our manual analysis, we made every effort to identify all data collected by the trackers. However, due to the complexity of the task, some information may have been overlooked.

Our study may have been unfeasible if a tracker had used reflection to obfuscate its code and dynamically download the strings used in the class and method calls. Fortunately, we did not encounter this problem in the analyzed trackers.

Our research was conducted under rigorous ethical procedures. To understand the behavior of the trackers and detect what information was being collected, we needed to perform reverse engineering steps of the app binaries (APK files). We limited the scope of these activities to identify what data from the users, their device, and their app usage were harvested and then sent to the tracker servers. No other action was done with the binaries.

All platforms we used, like *Google Play* and *Exodus* website, were accessed according to their terms of service. We analyzed every possible country, region, and jurisdiction and examined all available applications thoroughly.

4 FINDINGS

In this section, we put forward the results of our research. We provide details of the analyzed trackers and discuss the privacy issues caused by using those tools in implementing COVID-19-related applications.

4.1 Applications and Trackers Statistics

We have identified 58 different trackers present in the 595 apps analyzed. In Table 2, it can be seen that trackers detected by *Exodus* such as *Google Firebase Analytics* and *Google CrashLytics* are present in a significant number of the apps of the data set, especially the former (almost two out of three apps contained it). Notably, nearly half of the detected trackers, 28 out of 58, were included in no more than two apps.

Trackers	Applications	Percentage
Google Firebase Analytics	357	60.00%
Google CrashLytics	180	30.25%
Google AdMob	47	7.90%
Facebook Login	40	6.72%
Facebook Share	38	6.39%
Facebook Analytics	34	5.71%
Google Analytics	28	4.71%
OneSignal HMS Core	24	4.03%
Microsoft Visual Studio App Center Crashes	22	3.70%
Microsoft Visual Studio App Center Analytics	20	3.36%
Facebook Places	18	3.03%
Facebook Ads Google Tag Manager Amplitude	14	2.35%
OpenTelemetry	13	2.19%
AltBeacon Bugsnag	12	2.02%
Matomo	9	1.51%
Segment	8	1.35%
Mapbox Branch	7	1.18%
New Relic Braze (*)	5	0.84%
MixPanel	4	0.67%
Splunk MINT Pushwoosh Facebook Flipper Airship Flurry	3	0.50%
Esri ArcGIS Demdex AppMetrica Appcelerator Analytics AppsFlyer Adobe Experience Cloud HyperTrack Bugfender County	2	0.34%
Startapp AdColony Radius Networks RjFun Heap MOCA Kontakt ComScore LotaData Snowplow Pusher Appnext Split Instabug Conversant Scandit TNK Factory Bolts Lotame	1	0.17%

Table 2: Number of applications (out of 595) in which the identified trackers were found, together with their percentage of occurrence. (*) Braze was previously known as Appboy, which appears with this name in the binary packages. Same with ex Urban Airship now known as Airship.

Given their functionality and services, developers are encouraged to add SDKs to their apps. The trackers in our study provided the following services:

- **App Usage, Audience and Engagement:** analytics of how the app is used, and by who, events monitoring and tracking, user engagement (26 trackers);
- **Crash Reporting and Monitoring:** monitoring of resources, logs, crashes, and general app functioning (10 trackers);
- **Technical Functionality:** Specific functionality for the user that is integrated into the app (e.g., maps, beacons, and code scanning) (8 trackers);
- **Mobile Advertising:** distribution of ads into the app, monetization (5 trackers);
- **Devel. and Back-end Framework:** Libraries for ease of app development and back-end services (e.g., hosting, storage, and distribution) (3 trackers);

Main Service	Apps Count
Devel. and Back-end Framework	382
Crash Reporting and Monitoring	242
App Usage, Audience, and Engagement	156
Social Media Integration	96
Mobile Advertising	77
Push Notification	28
Technical Functionality	27

Table 3: For each service category, a count of times trackers belonging to these categories were included.

Trackers Count	Applications	Percentage
None	193	32.44%
One	123	20.67%
Two	160	26.89%
Three	49	8.24%
Four	22	3.70%
Five	18	3.03%
More than five	30	5.04%

Table 4: Number of trackers found in the studied applications, and their respective percentage.

- **Push Notification:** push notification and other messaging channels (3 trackers); and
- **Social Media Integration:** integration with social media within the app (e.g., login and content sharing) (3 trackers).

In Table 3, we show the number of times trackers in the categories described above were included in the studied apps. While it can be seen that libraries that assist in the development and maintenance of apps in production were broadly used, trackers offering advertisement services, whose pertinence with the studied apps is debatable, were included 77 times.

From the perspective of the apps, in Table 4, it can be appreciated that almost one-third of them do not include any tracker. Around half of them only contain one tracker and around 80% of the apps include no more than two trackers. On the other hand, 30 apps had over five trackers. In Table 5, we show the 15 apps that presented the most trackers in our data set, ranging from 11 to 9. The purpose of these apps varies; some have been downloaded hundreds of thousands or even millions of times.

Given the urgency to provide digital health apps to restrain the spread of the virus at the beginning of the pandemic, it would not be surprising if initially, applications extensively used third-party libraries and then removed or reduced their use in subsequent releases. Using historical information of application reports from *Exodus*, we queried the number of trackers detailed in each one of these reports and compare the number of trackers found between the first and the last available report, to determine whether the tracker number per application decreased over time. Our goal was to understand if, since the containment of the virus was successful, governments, organizations, and other providers of COVID-19-related apps improved their privacy aspects. The results can be seen in Table 6: for apps that contained more than one privacy report (153 in total), so there could be a comparison, 25 of them

Application	Region	Goal	Trcks.	Downls.
WebMD: Symptom Checker	International	Info.	11	10M+
COVID Symptom Study	UK	Studies	10	1M+
Korona Önlem	Turkey	Self-Diagn.	10	-
C Spire Health - UMMC Virtual COVID-19 Triage	USA	Info.	10	10K+
Sydney Care	USA	Info.	9	-
Manitoba Immunization Verifier	Canada	Cert. Verif.	9	50K+
Manitoba Immunization Card	Canada	Cert.	9	50K+
QMUNITY	Malaysia	Contact Tracing	9	-
Ministry of Health, Trinidad and Tobago	Trinidad and Tobago	Telehealth	9	1K+
CoVerified	USA	Cert.	9	10K+
SMART Health Card Verifier	USA	Cert. Verif.	9	100K+
SafeEntry (Business)	Singapore	Contact Tracing	9	100K+
Covid-19 Cuernavaca	Mexico	Info.	9	-
WHO LENA	International	Health-workers	9	1K+
RBC-C19	Rwanda	Cert.	9	10K+

Table 5: Top 15 applications with most trackers included. The count of downloads at the moment of writing this article is included, showing “-” if the application is no longer in Google Play.

Case	Status	Count
Decrease	Trackers at end	19
	No trackers at end	6
Increase	Trackers at start	15
	No trackers at start	6
Equal	Trackers	56
	No trackers	51
One report	-	184
No Report	-	258

Table 6: Increase and decrease of trackers per application, based on *Exodus* historical reports.

(≈16%) decreased the count of trackers included, with 6 of these 25 ending with no trackers detected in their latest report; 21 (≈14%) increased their count of trackers, with 6 of them starting with no trackers but including at least one at some point; and 107 apps (≈70%) maintaining the same number of trackers between their first and last report (56 of these including trackers in their binaries).

4.2 On Data Collection

Our main findings are that user, device, and usage data were tracked. The *modus operandi* of trackers is shown in Figure 2. The tracker’s

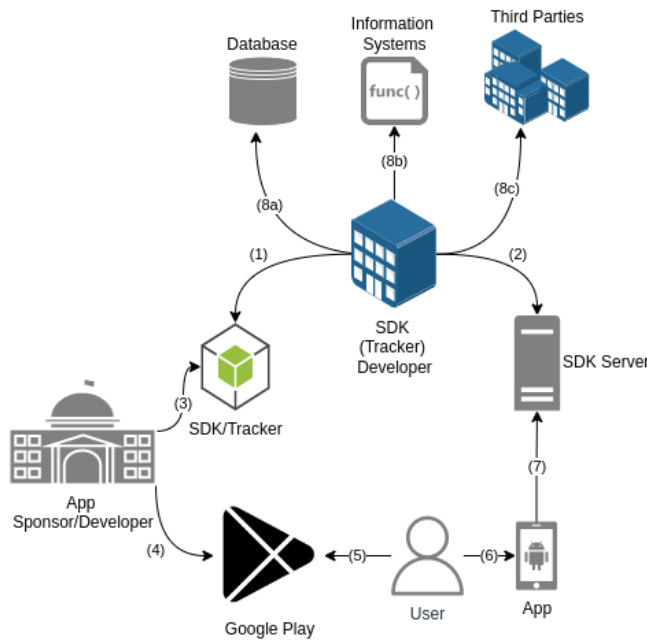


Figure 2: Tracker data collection example.

company offers mobile developers an SDK that provides specific functionality when integrated into the application (1). Aligned, they set up the SDK server so the tracker can connect back when in use (2). Later, a developer, in this case probably a government, international organization, or someone sponsored by them, will include the SDK into its COVID-19-related application (3), which will be sent to Google Play (4) for release. Mandated by their government, compelled by living circumstances, fearful of being misinformed, or by any other incentive, users will download these COVID-19 apps (5) and start using them (6). While running, at some point, the SDK code will run, including the part dedicated to tracking information, and, at a given time, it will upload the information harvested to the SDK servers (7). Finally, the company behind the SDK will make use of the collected data storing it (8a), for example, to fingerprint the device [15], processing it so they can profile the user of the application (8b) (*Engagement Data* in [14]), or simply sharing it with third parties for a revenue (8c) (point 9 in [18]).

To verify exactly what pieces of information the trackers harvest, we checked their code in the apps shown in Table 7. Since SDKs included in the apps may vary between releases, and even the code of the tracker may change over time, we also detail their corresponding versions.

Based on our findings, the data gathered by the trackers can be cataloged into the following categories:

- **Android Advertisement ID (AAID):** To this category, as the name suggests, belongs the *Android Advertisement ID* and whether its tracking is limited.
- **User ID:** Most identification artifacts in this category are based on *UUID* variants. The ADID or user ID was sometimes used directly from the main application code. In a few cases,

Tracker	Tracker Version	Application Package	App. Version
AdColony	4.1.2	covid19.cuernavaca	9.8
Airship	5.1.0	au.com.vodafone.dream-labapp	3.3.1.3218
AltBeacon	2.16.4	gov.georgia.novid20	1.0.467
Amplitude	2.23.2	sg.gov.tech.safeentry	0.11.0
AppNext	2.4.5.472	covid19.cuernavaca	9.8
Branch	3.2.0	ca.bc.gov.health.hlbc.COVID19	1.42.0
Braze	8.0.0	com.clearme.clearapp	1.11.1
Bugsnag	5.1.0	gov.adph.exposure-notifications	1.10.0
Flurry	11.5.0	mu.mt.healthapp	2.0.103
Google Ad-Mob	12.4.51	tr.gov.saglik.korona-onlem	1.0.3
Google Firebase Analytics	17.0.0	ca.ontario.verify	1.1.1
Google Tag Manager	5.06	az.gov.my	1.6.1
Mapbox	9.6.2	com.healthcarekw.app	2.1.9
Matomo	N/A	cy.gov.dmid.covtracer	3.3.12
MixPanel	4.8.7	com.moh.alert.ramzor	1.15.0
New Relic	6.3.1	ar.gob.coronavirus	3.5.32
OneSignal	3.12.3	uy.gub.salud.planacovid-19uy	9.1.1
Open Telemetry	0.21.0	es.gob.asistenciacovid19	1.0.11
	0.21.0	au.gov.health.covid19	1.4.10
Pushwoosh	6.3.6	gov.cdc.general	3.1.4
Segment	4.8.2	com.thecommons-project.smarthealthcard-verifier	1.0.27
Splunk MINT	5.0.0	et.gov.moh.oppia.covid	7.3.0-et.2.int
Startapp	4.5.0	covid19.cuernavaca	9.8

Table 7: Applications that were analyzed to detect tracking behavior. We studied two applications for *OpenTelemetry* to validate our findings. In the case of *Segment*, we found a pattern in applications developed using the *EXPO* framework, where several wrappers for trackers (including *Segment*) were automatically included in the applications.

an ID was assigned at installation time or was set with a push notification.

- **Location Software:** In this category, we include tracking information that can approximate the location of the application user, which is based on the software configurations of the device. For example, here we group calls to track the *locale*, *language*, and *timezone*, *country code*, *input languages* and *daylight saving*.
- **Location Hardware:** Contrary to the above, we grouped the calls to get the user's and device's exact geographic location in this category.
- **Device Software:** This category includes the information related to the operating system (its name, version, architecture, and build) and the *user-agent* used at connection time.

In general, the OS name was hardcoded as "Android" in the trackers' code.

- **Device Hardware:** The information about the physical device was grouped within this category, including elements like: the device *manufacturer*, *model*, and *brand*, *name*, *id*, *board*, *display* and if it is a *tablet* or *phone*. Also, in this category, we included the tracking of the *SIM card*.
- **Android Package Kit (APK):** Another typical piece of data tracked was the application's information, including the tracker SDK. Elements like the package *name*, *version name*, and *version code* belong to this category. Also, information related to the *application bundle*, the *permissions granted*, the *environment*, the development *framework*, if it is an *instaApp* [13], and the *app store*, and its given ID by the tracker on its platform.
- **Applications/Processes:** Some trackers gathered information about the *activities running*, the *installed apps*, the *state* of the application, the *threads running*, and the *Java Virtual Machine*.
- **Disk/Memory:** There were some trackers that gathered information about the *disk*, the *filesystem* and the *memory* of the device.
- **Network:** In this category, we put the calls the trackers did to get information about the *carrier*, the *network* used by the device, the *wifi*, and *bluetooth*. For a particular tracker (*Startapp*), we observed the collection of over one hundred different elements related to this category.
- **Screen/Audio:** Information about the *display* and its *orientation* was commonly harvested. There was also information related to the audio, like the *ringer mode*, if an *earplug connection* exists, and the *volume*, among others.
- **Rooted/Jailbroken/Emulated/Simulated:** A few trackers tried to identify if the device was *rooted/jailbroken* or if it was an *emulation/simulation*.
- **Time:** In this category, time-related tracking activities are grouped. For example, the time of the *application install*, its *last update*, or other *events* that the trackers monitor. An example that is not a timestamp is the *session duration*.
- **Battery:** Information about the battery was also tracked, namely its *level* and if it was being *charged*.
- **Software Development Kit (SDK):** The tracker tracked data of itself as well, for example, its *version* and *flavor* (hard-coding these values in their code), and
- **Others:** In addition to the categories described above, trackers harvested many other dissimilar types of information from the device and its usage; a clear example of this is the *metadata* and *type* associated with the events they monitored.

In Table 8, we detail what information was found to be harvested by each tracker in our analysis of their code. The data tracked is classified based on the categories previously described. *AltBeacon*, *Google Tag Manager* and *OpenTelemetry* are highlighted in green since we could not find tracking behavior in their code. On the other hand, *Google AdMob* and *Startapp* are painted in a red tone given that they track information in every described category.

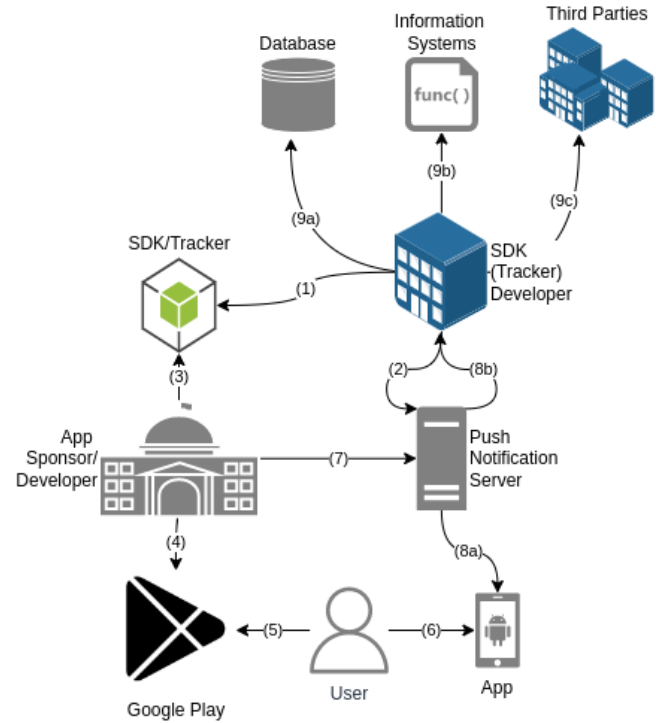


Figure 3: Tracker push notification example

4.3 On Push Notifications

Another problem we found was using push notifications to communicate sensitive information to the application users. Indeed, given the type of application we study, push notifications can include sensitive information, such as exposure alerts, COVID-19 test results, and health-related notifications, among others. Similarly to the data collection functionality, we illustrate this behavior in Figure 3. First, the SDK developer will provide its push notification SDK (1) and associated server (2). As in the previous case, the developer will include the push notification SDK in the application (3) and publish it in the store (4). The user will then download (5) and use the application (6). At the moment of sending a new push notification to all users, a specific group, or a targeted user of the application, the developers of the apps need to choose the recipients and specify the content of the message to be sent using the service platform at the push notification server (7). This means the message is passed in plaintext to the SDK provider, who can store, process, or transfer this information to a third party (8).

Out of the 22 trackers we analyzed, we found that 6 of them (*Airship*, *Braze*, *Google Firebase Analytics*, *Flurry*, *OneSignal* and *Pushwoosh*) included push notification services to the apps. We could not find any reference to end-to-end encryption in the services documentation, except for the case of *Pushwoosh* [3]. This shall be discussed in Section 5.

Other trackers found in our study that also provide push notification services are: *Adobe Experience Cloud*, *AppMetrica*, *Countly*, *Facebook Analytics*, *HMS Core*, *LotaData*, *MOCA*, and *Pusher*.

Tracker	AAID	User ID	Loc. SW	Loc. HW	Dev. SW	Dev. HW	APK	A/P.	D/M	Net.	S/A	R/E	Time	Batt.	SDK	Oth.
AdColony	✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓	✓
Airship		✓	✓	✓	✓	✓	✓	✓		✓	✓		✓		✓	✓
AltBeacon																
Amplitude	✓	✓	✓	✓	✓	✓	✓			✓			✓			✓
AppNext		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓
Branch	✓	✓	✓		✓	✓	✓			✓	✓		✓		✓	✓
Braze	✓	✓	✓	✓	✓	✓	✓			✓	✓		✓		✓	✓
Bugsnap		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Flurry	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓	✓
Google AdMob	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Google Firebase Analytics		✓			✓	✓	✓								✓	✓
Google Tag Manager																
Mapbox		✓		✓	✓	✓		✓		✓	✓		✓	✓	✓	✓
Matomo		✓	✓			✓	✓				✓		✓		✓	✓
MixPanel					✓	✓	✓			✓	✓				✓	
New Relic		✓	✓		✓	✓	✓	✓	✓	✓	✓				✓	✓
OneSignal	✓	✓	✓	✓	✓	✓	✓			✓		✓				✓
OpenTelemetry																
Pushwoosh		✓	✓		✓	✓	✓			✓		✓	✓		✓	✓
Segment	✓	✓	✓		✓	✓	✓			✓	✓				✓	✓
Splunk MINT		✓	✓		✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓
Startapp	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 8: Type of information harvested by the trackers and sent to their servers.

5 DISCUSSION

In this section, we elaborate on our findings and assess the impact of the problem from the users' perspective.

5.1 On the Known Practices of Device Identification: Impact of Data Collection

Identifying devices through fingerprinting and segmentation is a known practice. Researchers propose efficient fingerprinting mechanisms to improve authentication, while trackers openly state their data collection purpose for device identification and attribution. This is even easier if they collect the ADID of the phone.

The use of tracking and identification in COVID-19-related apps can pose a serious risk to users' sensitive health data. Out of the total of processed apps, 34 were primarily used for quarantine enforcement, and 29 of them contained trackers that collected the APK identifier along with other information. This means that if the companies behind these trackers knew that these apps were used for quarantine enforcement, they could infer that the user was likely infected with COVID-19. Furthermore, some trackers can collect location information with a high level of accuracy, potentially revealing where users with the device were infected. This information could also be applied to vaccination scheduling and COVID-19 certification apps, allowing third parties to track the user's movements and potentially identify the times they visit hospitals. Additionally, some trackers collect the SSID of Wi-Fi hotspots within range, allowing them to confirm the user's location.

5.2 A Design Issue: Impact of Using Push Notification Services

As emphasized in Section 4, developers are required to provide the push notification service provider with the message content of the push notification in plaintext. This means that messages such as "Your test results were positive..." or "You are scheduled for your X COVID-19 vaccination dose..." have the potential to reveal sensitive health information of users to third parties. What is worse is that application users are not informed nor have they given consent for this specific information disclosure to occur to third parties.

Ensuring end-to-end encryption for push notifications is crucial for safeguarding user privacy. Unfortunately, proper implementation of this feature is not always guaranteed. Typically, encryption occurs on the service provider's servers, which leaves messages vulnerable to being processed in plaintext before encryption. As a result, confidentiality may be compromised.

Developers should consider alternatives in the context of COVID-19 sensitive information being sent over push notifications. It is important to rethink the necessity of these notifications or consider alternatives. While an email or SMS may have similar risks, a phone call could be a safer option. Another possibility is to use push notifications solely as an alert, prompting users to open the application for more information. Alternatively, developers could opt for real end-to-end encryption libraries like *Capillary* [2], which offer secure communication between servers and devices.

5.3 Data Privacy Posture from Governments

Maintaining data privacy is crucial, but in the face of a pandemic, certain tracking mechanisms in apps may be necessary to curb the

spread of the virus. Nevertheless, it is alarming that the number of trackers in some apps did not diminish, and in some cases, even grew, even after the pandemic had been contained. Table 6 shows that the number of trackers included in some apps increased during the pandemic.

We investigated app privacy policies and found some governments were not transparent about trackers. However, some developers disclosed third-party libraries. For example, the app *Covid-19 DXB* declared *Google Play Services*, *Google Analytics for Firebase*, and *Firebase Crashlytics* [8]. The app *CovidSafe* may transfer data to *Firebase* with user permission, but not certificate content [7].

Some governments have taken measures to safeguard data privacy by conducting Data Protection Impact Assessments (DPIA) for COVID-19-related apps [9]. This is a significant advancement towards normalizing privacy-by-design approaches. Some have analyzed the inclusion of trackers in the app, ensuring that no third-party trackers collect personal data. Others have evaluated the impact of using tools like *Firebase* and *Microsoft SDK* for push notifications and performance diagnosis [11]. However, some DPIAs have overlooked the analysis of tracker impact [10].

It can be inferred that certain users were not fully aware of the privacy policies of the apps they were compelled to install, and that some of the apps were developed without fully analysing the privacy implications of including trackers.

5.4 What do the Trackers Declare?

We found that, generally, trackers publicly declare what data they collect, explicitly enumerating the collected items in their privacy policies. However, some claim fewer items than we found during our code analyses. The overall result of this study can be appreciated in Table 9. The list of analyzed policies can be found at our public project [20]. We estimate that Google’s requirement for their *Google Play’s new safety section* has encouraged (or, up to a point, mandated) trackers to transparently indicate the data they collect with their SDKs.

To the best of our efforts, we could not find a disclosure by *Amplitude* and *Segment* (Twilio) of the data they collect from their Android SDKs. No privacy policy or data collection information was found for *AltBeacon* and *Open Telemetry* (but we also did not find any tracking behavior from them).

Finally, some trackers, like *Startapp*, openly declare that they **will share/sell** the tracked information from the users of the apps where their SDK is included.

5.5 Research Questions Discussion

As stated in Section 1, the primary goal of this study was to answer three main questions related to COVID-19 applications. In what follows, we analyze each of them:

(RQ1) Are trackers being used in the COVID-19 app ecosystem? As described in Section 4, trackers were extensively used in COVID-19 mobile apps. We found a total of 58 different trackers in 595 apps, some of them containing over 5 trackers. However, it is worth noting that almost one-third of the apps in our data set did not include any tracker. In addition, our historical research of *Exodus* reports demonstrated that the number of trackers included in the apps did not necessarily decrease throughout the pandemic.

Tracker	Declares	Coincides
AdColony	Policy	No
Airship	Enumeration	Yes
AltBeacon	-	-
Amplitude	Policy	No Data
AppNext	Policy	No
Branch	Mix	Yes
Braze	Enumeration	No
Bugsnap	Enumeration	Yes
Flurry	Enumeration	Yes
Google AdMob	Enumeration	No
Google Firebase	Enumeration	Yes
Google Tag Manager	Enumeration	Yes
Mapbox	Policy	Yes
Matomo	Enumeration	Yes
MixPanel	Enumeration	Yes
New Relic	Enumeration	No
OneSignal	Enumeration	Yes
Open Telemetry	-	-
Pushwoosh	Enumeration	No
Segment	Policy	No Data
Splunk MINT	Enumeration	Yes
Startapp	Policy	No

Table 9: Trackers declaration of data collected with their Android SDK. Column *Declares* has the values *Enumeration* if they enumerate each element they track, *Policy* if they provide a general idea in their Privacy Policy, but without explicitly detailing the collected items, or *Mix* if they enumerate within the Privacy Policy. Column *Coincides* indicates if what they declare coincides with our findings.

(RQ2) What information is being tracked? Our analysis of the trackers’ code allowed us to precisely identify the information being collected and sent to their servers. We found that in COVID-19 related apps, information shared with third-parties included the *Android Advertisement ID* of the device; its specific *location* and *locale* configuration; its *operating system*, *model*, *brand*, *manufacturer* and *rooted* condition; the status of *memory*, *disk*, *screen*, *audio*, and *battery*; *network* data; and the *application* where the trackers were included, plus other *running apps and processes*. We contrasted these elements with what trackers detail in their data collection disclosure documentation and found that, in some cases, there was no disclosure of all items being collected. Furthermore, the use of push notification services also has the potential to disclose sensitive information related to the health status of app users to the providers of these services, depending on the content of the messages.

(RQ3) What are the potential impacts on users if trackers are in use and harvest that information? Section 5 discusses the possible results of including trackers and push notification services for COVID-19 apps. We conclude that, under certain conditions, sensitive information related to users’ health could be disclosed to third parties, potentially violating applicable regulations. For example, since trackers collect the *app ID*, in case this app was used to enforce quarantine or to track the disease evolution in a user, they could infer that the app user was infected with COVID-19.

Moreover, since they collect *location* information, they could identify where infected users were and the places they visited (including identifying in this manner their health provider). Another example of disclosure to third parties could be when the users' positive infection status is notified using push notification services. Additionally, some trackers openly disclose that data collected could be transferred or sold to other parties. Therefore, the impact on the users increases when these trackers are used. Finally, since trackers were included in apps with millions of downloads, the number of affected users is extremely large (hundreds of millions).

6 CONCLUSION AND FURTHER WORK

We analyzed the data leaks caused by trackers in mobile applications related to COVID-19 and discussed the results. Our investigation revealed that the information transferred to trackers' servers, either directly through push notifications or indirectly through harvested application and device data, could have allowed companies to infer the COVID-19 status of hundreds of millions of users. We also discussed data protection initiatives from the government sector, which provide guidelines for the privacy of citizens' data used by these applications. However, industry and government sectors seem largely unaware of these libraries' impact on data protection.

Moving forward, we plan to extend our research on trackers to other application ecosystems, such as financial and payments, where the violation of bank secrecy could occur. Additionally, we want to explore the health domain, where apps may allow trackers to collect user information without their consent. Further studies are necessary to identify why trackers are included in critical apps, which can encourage best practices for privacy analyses.

REFERENCES

- [1] 2013. Zenodo, CERN. www.eui.eu/Research/Library/ResearchGuides/Economics/Statistics/DataPortal/Zenodo. Accessed: 2022-09-20.
- [2] 2018. Capillary. <https://github.com/google/capillary>. Accessed: 2022-10-02.
- [3] 2019. Pushwoosh Push Notification Privacy. <https://blog.pushwoosh.com/blog/how-to-work-with-sensitive-data-if-you-want-to-use-push-notifications-2/>. Accessed: 2022-09-24.
- [4] 2020. Coronavirus disease COVID-19. www.who.int/health-topics/coronavirus. Accessed: 2023-05-09.
- [5] 2020. COVID-19 Contact tracing: data protection expectations on app development. <https://ico.org.uk/media/for-organisations/documents/2617676/ico-contact-tracing-recommendations.pdf>. Accessed: 2022-10-04.
- [6] 2020. COVID-19 Dashboard, Johns Hopkins University. www.arcgis.com/apps/dashboards/bda7594740fd40299423467b48e9ecf6. Accessed: 2023-05-09.
- [7] 2020. Covid Safe Privacy Policy. <https://cert-app.be/en/privacy-covidsafe.html>. Accessed: 2022-09-30.
- [8] 2020. COVID19 - DXB Smart App Privacy Policy. <https://www.dha.gov.ae/en/privacy-policy>. Accessed: 2022-09-30.
- [9] 2020. DPIA Covid Tracker App. <https://github.com/HSEIreland/covidtracker-documentation/blob/master/documentation/privacy/>. Accessed: 2022-09-30.
- [10] 2020. DPIA CovidCert. <https://covid-19.hscni.net/covidcert-ni-mobile-app/>. Accessed: 2022-09-30.
- [11] 2020. DPIA NHS. https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1028998/NHS_COVID_19_App_DPIA.pdf. Accessed: 2022-09-30.
- [12] 2020. MIT Technology Review Covid Tracing Tracker. <https://www.technologyreview.com/2020/12/23/1015557/covid-apps-contact-tracing-suspended-replaced-or-relaunched/>. Accessed: 2022-10-03.
- [13] 2021. Instant Apps. <https://developer.android.com/topic/google-play-instant>. Accessed: 2022-09-23.
- [14] 2022. Branch Privacy Policy. <https://branch.io/policies/privacy-policy/>. Accessed: 2022-09-29.
- [15] 2022. Device Fingerprinting. <https://blog.getsocial.im/device-fingerprinting-for-mobile-attribution/>. Accessed: 2022-10-03.
- [16] 2022. Exodus Standalone. <https://github.com/Exodus-Privacy/exodus-standalone>. Accessed: 2022-09-20.
- [17] 2022. jadx - Dex to Java decompiler. <https://github.com/skylot/jadx>. Accessed: 2022-09-19.
- [18] 2022. Startapp Privacy Policy. <https://www.start.io/policy/privacy-policy-site/>. Accessed: 2022-09-29.
- [19] 2023. Analyzed binaries. https://gitlab.fing.edu.uy/gsi/trackers-covid/-/blob/main/listado_apps.csv. Accessed: 2023-06-05.
- [20] 2023. Analyzed policies. <https://gitlab.fing.edu.uy/gsi/trackers-covid/-/blob/main/policies.csv>. Accessed: 2023-06-05.
- [21] AWO Agency. 2020. Report on the privacy risks of COVID-19 software. (2020).
- [22] Amal Awadalla Ali, Asma Hamid Elfadl, Maha Fawzy Abujazar, Sarah Aziz, Alaa Abd-Alrazaq, Zubair Shah, Samir Brahim, Belhaouari, Mowafa Househ, and Tanvir Alam. 2020. Contact Tracing Apps for COVID-19: Access Permission and User Adoption. In *2020 7th International Conference on Behavioural and Social Computing (BESC)*. 1–7. <https://doi.org/10.1109/BESC51023.2020.9348327>
- [23] Kevin Allix, Tegawendé F. Bissyandé, Jacques Klein, and Yves Le Traon. 2016. AndroZoo: Collecting Millions of Android Apps for the Research Community. In *Proceedings of the 13th International Conference on Mining Software Repositories (Austin, Texas) (MSR '16)*. ACM, New York, NY, USA, 468–471. <https://doi.org/10.1145/2901739.2903508>
- [24] Reuben Binns, Ulrik Lyngs, Max Van Kleek, Jun Zhao, Timothy Libert, and Nigel Shadbolt. 2018. Third Party Tracking in the Mobile Ecosystem. In *Proceedings of the 10th ACM Conference on Web Science (Amsterdam, Netherlands) (WebSci '18)*. Association for Computing Machinery, New York, NY, USA, 23–31. <https://doi.org/10.1145/3201064.3201089>
- [25] Davide Caputo, Francesco Pagano, Giovanni Bottino, Luca Verderame, and Alessio Merlo. 2022. You Can't Always Get What You Want: Towards User-Controlled Privacy on Android. *IEEE Transactions on Dependable and Secure Computing* (01 2022), 1–1. <https://doi.org/10.1109/TDSC.2022.3146020>
- [26] Hyunghoon Cho, Daphne Ippolito, and Yun William Yu. 2020. Contact Tracing Mobile Apps for COVID-19: Privacy Considerations and Related Trade-offs. *ArXiv abs/2003.11511* (2020).
- [27] Andrea Continella, Yanick Fratantonio, Martina Lindorfer, Alessandro Puccetti, Ali Zand, Christopher Kruegel, and Giovanni Vigna. 2017. Obfuscation-Resilient Privacy Leak Detection for Mobile Apps Through Differential Analysis. In *Proceedings of the ISOC Network and Distributed System Security Symposium (NDSS)*. San Diego, CA.
- [28] Paul-Olivier Dehay and Joel Reardon. 2020. Proximity Tracing in an Ecosystem of Surveillance Capitalism. In *Proceedings of the 19th Workshop on Privacy in the Electronic Society (Virtual Event, USA) (WPES'20)*. Association for Computing Machinery, New York, NY, USA, 191–203. <https://doi.org/10.1145/3411497.3420219>
- [29] Majid Hatamian, Samuel Wairimu, Nurul Momen, and Lothar Fritsch. 2021. A Privacy and Security Analysis of Early-Deployed COVID-19 Contact Tracing Android Apps. *Empirical Softw. Engg.* 26, 3 (may 2021), 51 pages. <https://doi.org/10.1007/s10664-020-09934-4>
- [30] Yongzhong He, Xuejun Yang, Binghui Hu, and Wei Wang. 2019. Dynamic privacy leakage analysis of Android third-party libraries. *Journal of Information Security and Applications* 46 (2019), 259–270. <https://doi.org/10.1016/j.jisa.2019.03.014>
- [31] Kollnig Konrad, Binns Reuben, Dewitte Pierre, Max Van Kleek, Wang Ge, Omeiza Daniel, Webb Helena, and ShadboltNigel. 2021. A Fait Accompli? An Empirical Study into the Absence of Consent to Third-Party Tracking in Android Apps.
- [32] Xing Liu, Jiqiang Liu, Sencun Zhu, Wei Wang, and Xiangliang Zhang. 2020. Privacy Risk Analysis and Mitigation of Analytics Libraries in the Android Ecosystem. *IEEE Transactions on Mobile Computing* 19, 5 (2020), 1184–1199. <https://doi.org/10.1109/TMC.2019.2903186>
- [33] Azad MA, Arshad J, Akmal SMA, Riaz F, Abdullah S, Imran M, and Ahmad F. 2020. A First Look at Privacy Analysis of COVID-19 Contact-Tracing Mobile Applications. *IEEE Internet Things J* (2020).
- [34] Sergio Nesmachnow and Santiago Iturriaga. 2019. Cluster-UY: Collaborative Scientific High Performance Computing in Uruguay. In *Supercomputing*, Moisés Torres and Jaime Klapp (Eds.). Springer International Publishing, Cham, 188–202.
- [35] Abbas Razaghpanah, Rishab Nithyanand, Narseo Vallina-Rodriguez, Srikanth Sundaresan, Mark Allman, Christian Kreibich, and Phillipa Gill. 2018. Apps, Trackers, Privacy, and Regulators: A Global Study of the Mobile Tracking Ecosystem. In *NDSS*.
- [36] Ryan Stevens, Clint Gible, Jonathan Crussell, Jeremy Erickson, and Hao Chen. 2012. Investigating User Privacy in Android Ad Libraries.
- [37] Gioacchino Tangari, Muhammad Ikram, Kiran Ijaz, Dali Kaafar, and Shlomo Berkovsky. 2021. Mobile health and privacy: Cross sectional study. *BMJ* 373 (06 2021), n1248. <https://doi.org/10.1136/bmj.n1248>
- [38] Kouliaridis V, Kambourakis G, Chatzoglou E, Geneiatakis D, and Wang H. 2021. Dissecting contact tracing apps in the Android platform. *PLoS One* (2021).
- [39] Liu Wang, Ren He, Haoyu Wang, Pengcheng Xia, Yuanchun Li, Lei Wu, Yajin Zhou, Xiapu Luo, Yulei Sui, Yao Guo, and Guoai Xu. 2021. Beyond the Virus: A First Look at Coronavirus-themed Mobile Malware. *arXiv:2005.14619 [cs.CR]*
- [40] Haohuang Wen, Qingchuan Zhao, Zhiqiang Lin, Dong Xuan, and Ness Shroff. 2020. A Study of the Privacy of COVID-19 Contact Tracing Apps. In *Security and Privacy in Communication Networks*, Noseong Park, Kun Sun, Sara Foresti, Kevin Butler, and Nitesh Saxena (Eds.). Springer International Publishing, Cham.